Success Factors in Search Engine Optimization: A Quantitative Study of Technology Firms' Adaptability to Algorithm Changes

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Abstract—Search engines play a central role in how organisations are discovered online, yet constant algorithm changes create uncertainty around how search performance is produced and maintained. This study addresses the gap in organisational adaptability research by applying dynamic capability theory to examine how organisational processes support adaptability in environments characterised by algorithmic volatility. Drawing on a quantitative research design, data were collected from 81 SEO and digital marketing professionals working in technology-driven organisations. Multiple regression analysis tested seven SEO capability dimensions—algorithm monitoring, audit readiness, technical capability, content usability, content strategy, keyword strategy, and link building—as predictors of organisational adaptability. Findings show that only content strategy quality ($\beta = .373$, $p = .001$) and keyword strategy ($\beta = .249$, $p = .010$) significantly influence adaptability. Organisations that systematically refresh content, align it with user intent, and manage keywords strategically demonstrate stronger recovery and stability after algorithm updates. These results suggest adaptability in SEO is less about technical compliance and more about developing repeatable, learning-oriented strategic routines.

Index Terms—search engine optimisation, algorithm adaptability, dynamic capabilities, content strategy, keyword strategy, technology firms, digital marketing.

I. INTRODUCTION

Search engines shape how organisations are discovered and evaluated online. Websites ranking higher on search engine results pages (SERPs) gain greater visibility and user attention, particularly in information-rich environments [1]. Search visibility therefore represents a strategic resource rather than a purely technical outcome.

Early SEO research assumed relatively stable ranking environments where optimisation practices produce predictable results. This assumption is increasingly difficult to sustain. Algorithm adjustments can affect firms' online visibility and competitive position [2], creating decision-making uncertainty because ranking criteria change without full public explanation [3].

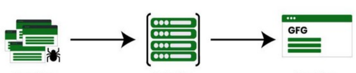

Fig. 1. Search Engine Workflow.

These challenges are particularly acute for technology-driven firms that rely heavily on organic search for product discovery. When search rankings change unexpectedly, firms may experience not only short-term drops in traffic but also longer-term consequences for credibility and market position. Despite this reliance on organic search, SEO is still often approached as a collection of technical tasks rather than as an organisational capability that develops over time.

This paper examines how organisational, operational, and strategic SEO practices shape technology firms' ability to adapt to search algorithm changes, drawing on dynamic capability theory, information processing theory, and absorptive capacity theory.

II. THEORETICAL BACKGROUND

A. Dynamic Capability Theory

Dynamic capability theory argues that firms in volatile environments achieve sustained performance by developing capabilities to sense environmental change, respond to opportunities, and reconfigure internal resources [4]. Algorithm updates represent recurring environmental shifts rather than isolated shocks. Firms that systematically monitor algorithm changes are better positioned to recognise shifts early, while firms with established response routines adjust practices in a timely manner [5].

B. Organisational Information Processing Theory

Organisational Information Processing Theory explains how organisations manage uncertainty through information collection, interpretation, and communication [6]. Search algorithm updates create uncertainty because ranking criteria change without full explanation. Organisations that draw on multiple information sources and use structured analysis deal with uncertainty more effectively [7].

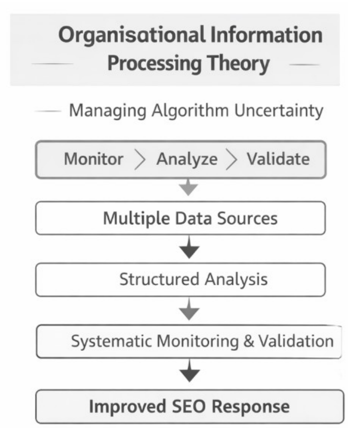

Fig. 2. Organisational Information Processing Theory in SEO Adaptability.

C. Absorptive Capacity

Absorptive capacity theory explains how organisations recognise useful external knowledge, interpret it, and apply it to improve performance [8]. Organisations with stronger absorptive capacity are more able to analyse algorithm changes, test different responses, and use what they learn to guide future practice. Structured experimentation and internal knowledge sharing support learning in rapidly changing environments [9].

D. Strategic Alignment

Strategic alignment highlights the need for fit between business strategy, information systems strategy, and day-to-day operations [10]. When content strategies, keyword planning, and link-building practices are coordinated with broader organisational goals, responses to algorithm updates are more likely to be systematic and consistent. Where alignment is weak, responses may become reactive and fragmented [11].

III. CONCEPTUAL FRAMEWORK AND HYPOTHESES

The conceptual framework proposes that organisational adaptability emerges from a combination of monitoring, learning, technical capability, and strategic alignment. Fig. 3 illustrates how these constructs collectively influence adaptability to algorithm changes.

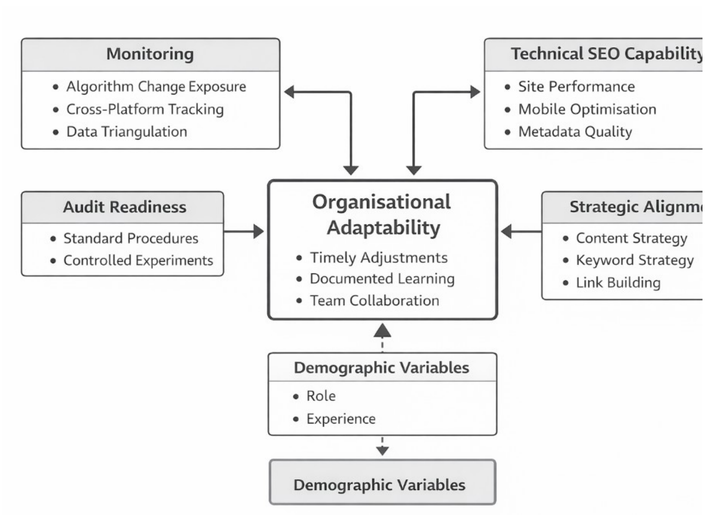

Fig. 3. Conceptual Framework of SEO Adaptability to Algorithm Changes.

Seven research hypotheses were derived:

H1: Algorithm change monitoring practices are positively associated with organisational SEO adaptability.

H2: SEO audit readiness and experimentation practices are positively associated with adaptability.

H3: Technical SEO capability is positively associated with adaptability.

H4: Content findability and usability are positively associated with adaptability.

H5: Content strategy quality is positively associated with adaptability.

H6: Keyword strategy is positively associated with adaptability.

H7: Link building is positively associated with adaptability.

IV. RESEARCH METHODOLOGY

A. Research Design

This study adopts a positivist, deductive, and quantitative cross-sectional design. A survey strategy was selected because it enables the collection of standardised data from a geographically dispersed population of SEO professionals. The questionnaire was developed directly from the conceptual framework, ensuring tight alignment between theory, hypotheses, and measurement.

B. Participants and Sampling

The target population consisted of professionals working directly with SEO in technology-driven organisations, including software/SaaS firms, e-commerce businesses, and technology-focused agencies. Purposive sampling was employed, supported by snowball recruitment through LinkedIn, SEO communities, and professional networks. Eligibility required a minimum of one year of SEO-related experience. A total of 81 valid responses were obtained.

C. Measurement

All constructs were measured using five-point Likert scales (1 = Strongly Disagree to 5 = Strongly Agree). The questionnaire comprised nine sections: demographics, algorithm monitoring, audit readiness, technical capability, content findability, content strategy, keyword strategy, link building, and organisational adaptability. Composite scores were created for each construct after internal consistency testing.

D. Data Analysis

Data were analysed using SPSS. Cronbach's alpha assessed scale reliability. Descriptive statistics, Pearson correlation analysis, and multiple linear regression were used to examine relationships and test hypotheses.

V. RESULTS

A. Sample Profile

Table I presents the distribution of professional roles. Web/technical SEO specialists represented the largest group (29.6%), followed by content strategists (27.2%) and digital marketing managers (22.2%). SEO analysts accounted for 19.8% and brand strategists 1.2%.

TABLE I
DISTRIBUTION OF RESPONDENT ROLES

Role	Freq.	Pct (%)
Brand Strategist	1	1.2
Content Strategist	22	27.2
Digital Marketing Manager	18	22.2
SEO Specialist / Analyst	16	19.8
Web Dev / Technical SEO	24	29.6
Total	81	100

Table II shows that 35.8% of respondents had 4–6 years of experience, 27.2% had 1–3 years, 27.2% had 7–10 years, and 9.9% had more than ten years of experience.

TABLE II
YEARS OF PROFESSIONAL EXPERIENCE

Experience	Freq.	Pct (%)
1–3 years	22	27.2
4–6 years	29	35.8
7–10 years	22	27.2
More than 10 years	8	9.9
Total	81	100

B. Reliability Analysis

All constructs demonstrated acceptable reliability. Cronbach's alpha values were: algorithm monitoring (α = .66), audit readiness (α = .71), technical SEO capability (α = .62), content findability and usability (α = .70), content strategy quality (α = .61), keyword strategy (α = .62), link building (α = .82), and organisational adaptability (α = .66). All values exceeded the .60 threshold commonly accepted in exploratory research [12].

C. Descriptive Statistics

Algorithm monitoring yielded the highest mean (M = 4.62, SD = 0.46), indicating widely embedded monitoring practices. Other constructs showed moderately high means: technical SEO capability (M = 3.84), content strategy quality (M = 3.80), keyword strategy (M = 3.80), link building (M = 3.76), organisational adaptability (M = 3.74), audit readiness (M = 3.70), and content findability (M = 3.67).

D. Correlation Analysis

Table III presents Pearson correlation coefficients between SEO capability constructs and organisational adaptability. Content strategy quality showed the strongest relationship (r = .616, $p < .001$), followed by link building (r = .574, $p < .001$) and keyword strategy (r = .510, $p < .001$). Technical SEO capability (r = .346, p = .002) and content findability (r = .402, $p < .001$) also showed significant positive associations. Notably, algorithm monitoring did not show a statistically significant relationship with adaptability (r = .065, p = .564).

TABLE III
PEARSON CORRELATION COEFFICIENTS (ORGANISATIONAL ADAPTABILITY)

Construct	r	p
Algorithm Monitoring	.065	.564
Audit Readiness	.288	.009**
Technical SEO	.346	.002**
Content Findability	.402	<.001***
Content Strategy	.616	<.001***
Keyword Strategy	.510	<.001***
Link Building	.574	<.001***

Note: ** $p < .01$; *** $p < .001$

E. Multiple Regression Analysis

A multiple linear regression model was estimated with all seven SEO capability constructs as simultaneous predictors of organisational adaptability. The overall model was statistically significant ($F(7,73) = 11.11$, $p < .001$), explaining 51.6% of variance in adaptability ($R^2 = .516$; adjusted $R^2 = .469$).

Two constructs emerged as significant predictors: content strategy quality (β = .373, p = .001) and keyword strategy (β = .249, p = .010). Remaining predictors—algorithm monitoring, audit readiness, technical capability, content findability, and link building—were not statistically significant in the full model.

TABLE IV
SUMMARY OF HYPOTHESIS TESTING RESULTS

H	Construct	Result
H1	Algorithm Monitoring	Rejected
H2	Audit Readiness	Rejected
H3	Technical SEO Capability	Rejected
H4	Content Findability & Usability	Rejected
H5	Content Strategy Quality	**Accepted**
H6	Keyword Strategy	**Accepted**
H7	Link Building	Rejected

VI. DISCUSSION

A. Monitoring and Audit Readiness

Despite high monitoring levels (M = 4.62), algorithm monitoring alone did not significantly predict adaptability in the regression model. This reflects findings in Information Processing Theory: collecting data is insufficient unless organisations have structures to interpret and act on it [6]. The correlation between audit readiness and adaptability (r = .288, p = .009) suggests that structured diagnostic procedures improve response quality, consistent with absorptive capacity theory [8].

B. Technical Capability

Technical SEO capability demonstrated a significant bivariate correlation with adaptability (r = .346, p = .002) but was not a significant independent predictor in the full model. This suggests technical optimisation functions as an enabling condition rather than a standalone driver—organisations need strong technical foundations to implement corrective actions quickly, but technical optimisation alone does not explain differential adaptation outcomes.

C. Content and Keyword Strategy as Primary Drivers

Content strategy quality (β = .373, p = .001) and keyword strategy (β = .249, p = .010) were the only significant predictors of adaptability in the full model. Organisations that systematically refresh content aligned with user intent and that manage keyword portfolios strategically demonstrate stronger recovery after algorithm updates.

These findings extend strategic alignment theory [10] to the SEO domain. Content and keyword strategies represent organisational practices that embed user needs into ongoing processes, creating natural resilience against algorithmic changes that themselves often target user experience improvements.

D. Link Building

Although link building showed a strong bivariate association with adaptability ($r = .574$, $p < .001$), it did not remain significant in the multivariate model, likely due to collinearity with content strategy. Link-building quality may therefore operate through strategic alignment rather than independently.

E. Theoretical Contributions

This study extends dynamic capability theory to SEO by demonstrating that adaptability can be operationalised and measured through organisational constructs. The finding that monitoring does not independently predict adaptability challenges a common practitioner assumption and suggests the importance of downstream learning and strategic processes.

VII. PRACTICAL IMPLICATIONS

Technology firms seeking to improve SEO resilience should prioritise: (1) developing systematic, intent-driven content strategies with regular refresh cycles; (2) implementing structured keyword planning across branded, long-tail, and localised terms; (3) establishing SEO audit protocols that translate monitoring data into actionable responses; and (4) strengthening cross-functional collaboration between SEO, content, product, and technical teams.

Rather than reacting to every algorithm change, organisations can build resilience by institutionalising content planning and keyword management as strategic organisational routines.

VIII. LIMITATIONS AND FUTURE RESEARCH

This study has several limitations. The sample ($n = 81$) limits generalisability, and the cross-sectional design captures perceptions at one point in time, precluding causal inference. Data relied on self-reported perceptions rather than objective performance metrics. The sample focused on technology-driven sectors, and results may differ across industries.

Future research should: examine larger, multi-sector samples; adopt longitudinal designs to track how adaptability develops over time; triangulate survey data with objective ranking and traffic metrics; and explore qualitative dimensions of SEO strategy through mixed-method designs.

IX. CONCLUSION

This study investigated how organisational, operational, and strategic SEO practices influence technology firms' adaptability to search engine algorithm changes. Drawing on a quantitative survey of 81 SEO professionals and applying dynamic capability, information processing, and absorptive capacity theories, the study found that adaptability is not driven by monitoring or technical optimisation alone.

Multiple regression analysis explained 51.6% of variance in adaptability. Only content strategy quality and keyword strategy emerged as significant independent predictors, emphasising that SEO resilience depends on strategic, learning-oriented practices rather than technical compliance. These findings advance understanding of SEO as an organisational capability and offer actionable guidance for managers navigating algorithmic volatility.

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